

6.4 Muons

6.4.1 Reconstruction

Muons are reconstructed using tracks in the ID and MS, together with their energy deposits in the calorimeters [1]. The main reconstruction categories are:

- **Combined muons** are reconstructed by matching and fitting ID and MS tracks, including the energy loss between the two tracking systems.
- **Segment-tagged muons** are reconstructed by matching an ID track to one or more local track segments in the MS when a complete MS track is unavailable.
- **Calorimeter-tagged muons** are reconstructed by matching an ID track to calorimeter deposits compatible with a minimum-ionising particle. Acceptance is recovered in regions with limited MS coverage.
- **MS-extrapolated muons** are reconstructed by extrapolating an MS track towards the interaction region without a matched ID track.

Combined muons are used in this analysis.

6.4.2 Identification

Muon identification tests whether the detector hits and the ID and MS momentum measurements are consistent with a single muon trajectory [1]. Decays of charged mesons in flight can change the trajectory and momentum, leading to a disagreement between the momentum measurements. For muon candidates, this agreement is quantified by the q/p significance,

$$S_{q/p} = \frac{|(q/p)_{\text{ID}} - (q/p)_{\text{MS}}|}{\sqrt{\sigma_{\text{ID}}^2 + \sigma_{\text{MS}}^2}}, \quad (6.2)$$

where the two q/p measurements are evaluated at the interaction point, and σ_{ID} and σ_{MS} are their uncertainties. Large values indicate poor agreement and help reject decay-in-flight or incorrectly matched candidates. Baseline muons are required to pass Loose identification. Signal muons must additionally satisfy the Medium working point, which imposes more restrictive requirements on the reconstruction quality.

6.4.3 Isolation

Muons from heavy-flavour decays are often accompanied by other particles inside a jet. They are suppressed using track and calorimeter isolation, defined by the surrounding activity after excluding the muon's own contribution [1]. The Loose_VarRad and Tight_VarRad working points combine track and calorimeter isolation with a track-cone radius that decreases as the muon p_T increases. Less surrounding activity is allowed by Tight_VarRad, giving stronger rejection of muons produced inside jets. Baseline muons are required to pass Loose_VarRad isolation, and signal muons must additionally satisfy Tight_VarRad.

6.4.4 Calibration

The muon momentum scale and resolution are calibrated using resonance decays, particularly $Z \rightarrow \mu\mu$ and $J/\psi \rightarrow \mu\mu$ [2]. The reconstructed mass distributions constrain biases from the magnetic-field description, detector alignment and material modelling. The corrections are applied to the combined momentum measurement used in the analysis. Reconstruction, identification and isolation scale factors are measured with tag-and-probe samples and applied to simulated event weights for the selected working points [1].

References

- [1] ATLAS Collaboration. Muon reconstruction and identification efficiency in ATLAS using the full run 2 pp collision data set at $\sqrt{s} = 13$ TeV. *Eur. Phys. J. C*, 81:578, 2021. doi: 10.1140/epjc/s10052-021-09233-2.
- [2] ATLAS Collaboration. Studies of the muon momentum calibration and performance of the ATLAS detector with pp collisions at $\sqrt{s} = 13$ TeV. *Eur. Phys. J. C*, 83:686, 2023. doi: 10.1140/epjc/s10052-023-11584-x.